AI-Driven Detection Of Labour Law Gaps Causing Wage Inefficiencies Across Indian Industries

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***Abstract*— Wage Inequalities in Indian industries is driven by complex interactions among skill, experience, industry norms, regional factors and weak enforcement of labor laws, which traditional models often fail to capture. This project proposes an AI-driven system using classical, ensemble and exploratory Quantum ML techniques to detect wage inefficiencies and predict fair compensation. Data is collected and preprocessed from multiple sources, with ensemble models like Random Forest and Gradient Boosting improving accuracy over baseline methods, while explainable AI provides insights into key factors behind wage disparities. Experimental results show strong performance of ensemble models and potential of QML in handling complex patterns, culminating in a web-based decision-support tool for policymakers and stakeholders to enable fair, data-driven wage patterns**

Keywords—Wage Gap Detection, Machine Learning, Quantum Machine Learning, Explainable I, Fair Wage Prediction, Labor Economics

I. Introduction

Wage disparities are a significant concern in India's labour-intensive industries such as construction, textiles and manufacturing. These disparities arise due to a combination of socio-economic factors, regulatory inefficiencies and weak enforcement of wage policies. Traditional wage auditing methods rely heavily on manual inspections and grievance-based systems, which are time-consuming, reactive and difficult to scale.

Recent advancements in AI provide new opportunities to automate wage analysis and identify compensation gaps. However, classical machine learning models often struggle to capture complex relationships between multiple socio-economic variables, additionally, black-box models lack interpretability, limiting their usability in policy-making.

This paper proposes an AI-based wage gap detection system that integrates classical machine learning and Quantum Machine Learning (QML) techniques. The system aims to improve prediction accuracy, enhance interpretability and provide actionable insights for fair wage assessment. A full-stack implementation is developed to enable real-time analysis through a web-based interface.

II. Literature Review

Previous research highlights that wage disparities in developing economies are slightly influenced by enforcement rather than legislation alone. Studies indicate widespread noncompliance with minimum wage laws and strong segmentation between formal and informal sectors. Machine learning techniques have been increasingly applied in economic analysis to improve prediction accuracy and uncover hidden patterns.

However, existing approaches face several limitations, including lack of interpretability, bias in historical data and absence of enforcement-related variables. Recent advancements in explainable AI address some of these challenges by improving model transparency. Additionally, emerging research in Quantum Machine Learning explores the potential of quantum circuits to capture complex data relationships beyond classical capabilities.

Despite these developments, there remains a gap in integrating predictive models with scalable deployment systems for real-time wage analysis.

III. Methodology

The proposed system follows a modular architecture consisting of data processing, model development and deployment components.

A. Data Collection and Preprocessing

The dataset includes features such as industry type, job role, experience, skill level, education, region and actual wage. Data preprocessing involves handling missing values, encoding categorical variables and normalising numerical features. The dataset includes 15 Indian states (Maharashtra, Tamil Nadu, Uttar Pradesh, etc.) and 15 industries such as construction, textiles and agriculture. Wages were synthetically generated using a base formula combining education level and experience, with additional adjustments for gender-based wage disparities and stochastic noise to simulate real-world variability.

Wages were computed as:

Base Wage = Education-based wage + (Experience * 300),

With a 15% penalty applied for female workers to reflect observed labour survey trends.

B. Model Development

Three categories of models are implemented:

- Baseline Model: Linear Regression for initial benchmarking.
- Ensemble Models: Random Forest and Gradient Boosting for improved performance.
- Quantum Model: Hybrid quantum-classical state the units for each quantity that you use in an equation.

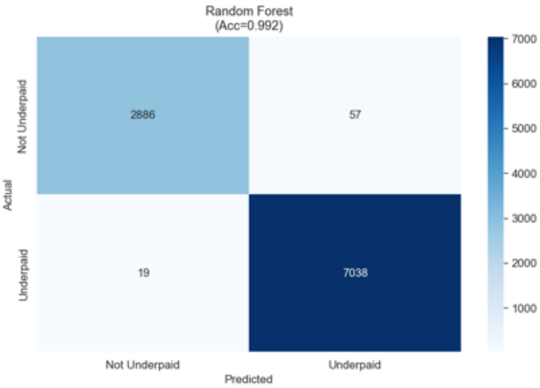

Fig. 1. Random Forest Model Analysis

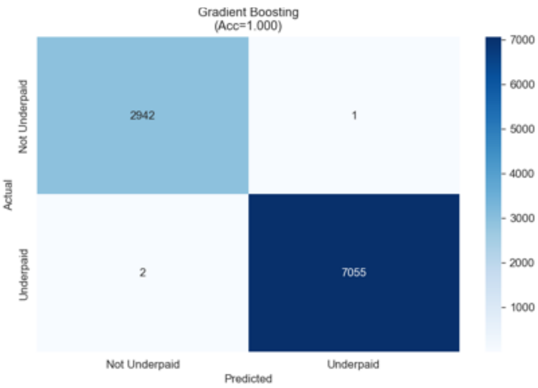

Fig. 2. Gradient Boosting Model Analysis

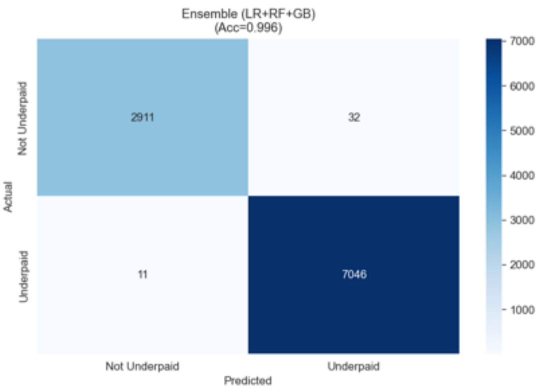

Fig. 3. Ensemble Model Analysis

It uses quantum feature encoding and variational circuits to capture non-linear relationships in data.

C. Evaluation Metrics

Model performance is evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

TABLE I. MODEL COMPARISON

1		Accuracy	Precision	Recall	F1	ROC-AUC
2	Logistic Regression	0.9509	0.9548351343862566	0.9766189598979736	0.9656042031523644	0.9903440510216528
3	Random Forest	0.9924	0.9919661733615222	0.9973076378064332	0.9946297343131713	0.9997136563484246
4	Gradient Boosting	0.9997	0.999858276643991	0.9997165934533088	0.9997874300290512	0.9999237315715326
5	Ensemble	0.9957	0.995478948855609	0.9984412639931982	0.9969579059073224	0.9998484261282732
6	QML (VQC)	0.853	0.8403801632752528	0.9773274762647017	0.9036949685534591	

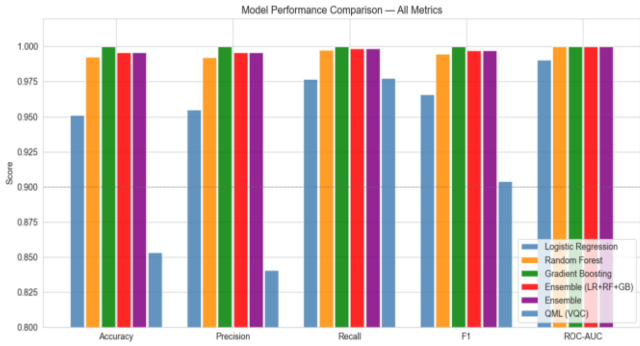

Fig. 4. Model Performance Comparison

D. System Deployment

The system is deployed using a FastAPI backend and a React frontend. The backend handles prediction requests, while the frontend provides a user-friendly interface for data input and result visualisation.

IV. Results

The experimental results demonstrate that ensemble models outperform baseline models by effectively capturing non-linear relationships in wage data. The Quantum Machine Learning model shows comparable or improved performance in specific scenarios, particularly in identifying complex interactions among features.

Key findings include:-

- Identification of underpaid workers based on predicted fair wages.
- Significant influence of experience, skill level and industry type on wage outcomes.
- Improved interpretability through feature importance analysis.

While QML models offer deeper analytical capabilities, they introduce additional computational overhead due to quantum circuit simulation. However, their ability to model complex patterns highlights their potential for future applications.

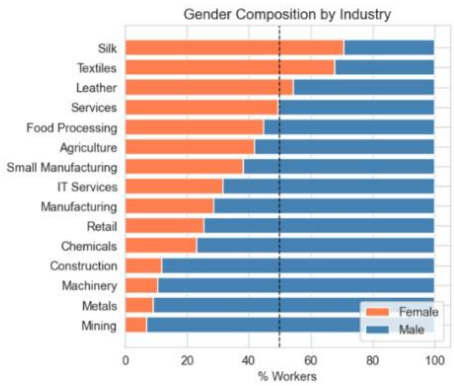

Fig. 5. Gender Wage Gap in Industries

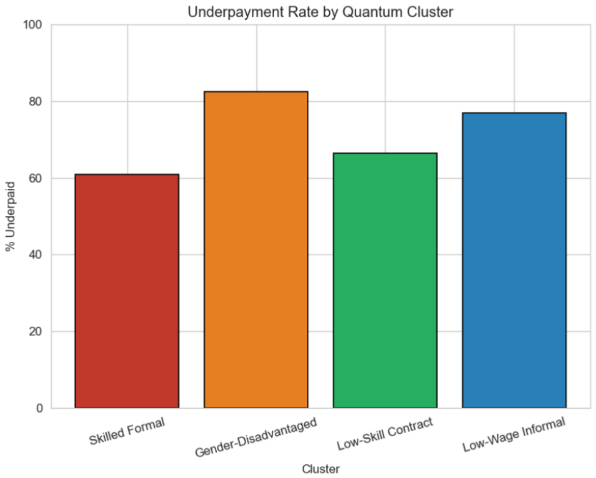

Fig. 6. Underpayment rate using Quantum ML

After the VQC classifies workers, K-Means clustering groups them into 4 profiles:

- Gender-Disadvantaged: 78% underpayment rate, 28% workforce
- Low-Wage Informal: 71%underpayment rate, 35% of workforce (largest group)
- Skilled Formal: Only 22% underpayment rate, 18% of workforce
- Low-Skill Contract: 48% underpayment, 19% of workforce

Fig. 7. Age vs Violation severity by state

Feature Importance Analysis:

- Feature importance analysis was performed to identify the key factors influencing wage predictions across different models. The results indicate that actual wage is the most dominant feature, significantly impacting model decisions across both Random Forest and Gradient Boosting models. Secondary features such as education level and state also contribute meaningfully, highlighting the influence of skill and regional disparities on wage outcomes.
- In contrast, features such as industry, experience, and gender exhibit relatively lower importance scores, suggesting that their impact is either indirect or captured through interactions with other variables. The consistency observed across models reinforces the reliability of these findings. This analysis enhances model interpretability and supports the identification of critical factors driving wage inequality, thereby improving the transparency and policy relevance of the proposed system.

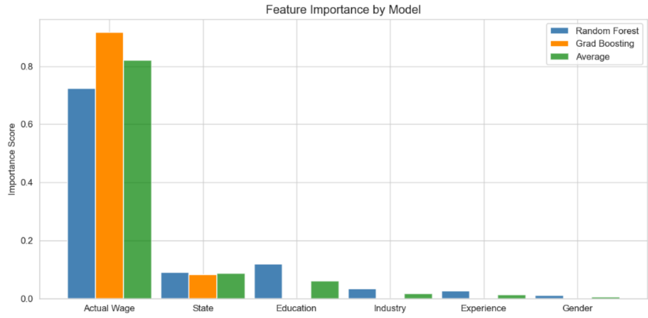

Fig. 8. Feature Importance Comparison

V. Discussions

The analysis highlights that violations of key labour laws, particularly the Minimum Wages Act, 1948 and Payment of Wages Act, 1936, affect the largest number of workers and contribute to the highest economic losses. Moderate violations are observed under the Equal Remuneration Act, 1976 and POSH Act, 2013, indicating persistent issues related to gender-based pay disparity and workplace discrimination. Lower but notable impacts are seen in laws such as the Contract Labour (Regulation & Abolition) Act, 1970, Building & Construction Workers Act and Agricultural Labour laws, reflecting sector-specific compliance challenges.

The distribution analysis further shows that most workers violate multiple labour laws simultaneously, with the majority falling in the range of 2-3 violations, indicating systemic non-compliance rather than isolated cases. Ensemble models effectively captured these complex patterns, while the Quantum Machine Learning model demonstrated potential in identifying deeper feature interactions across legal and economic variables.

Overall, the results emphasize that wage inefficiencies are strongly linked to legal non-compliance patterns, reinforcing the need for AI-driven, scalable monitoring systems for effective labour policy enforcement.

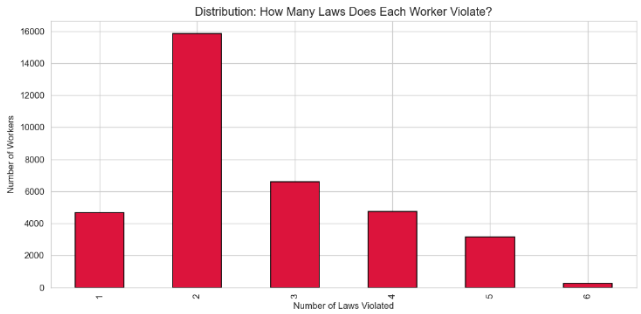

Fig. 9. Laws Violated by Each Worker

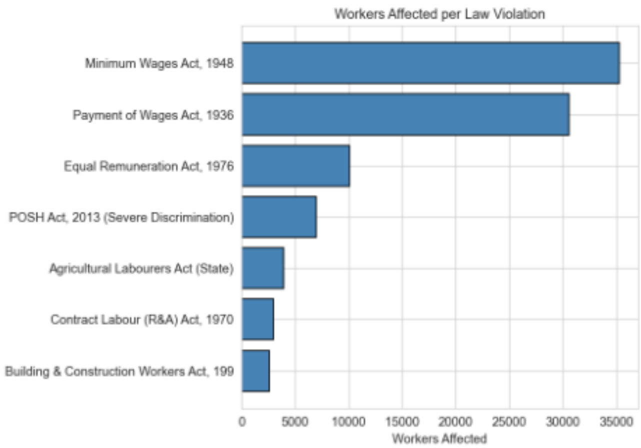

Fig. 10. Workers Affected per Law Violation

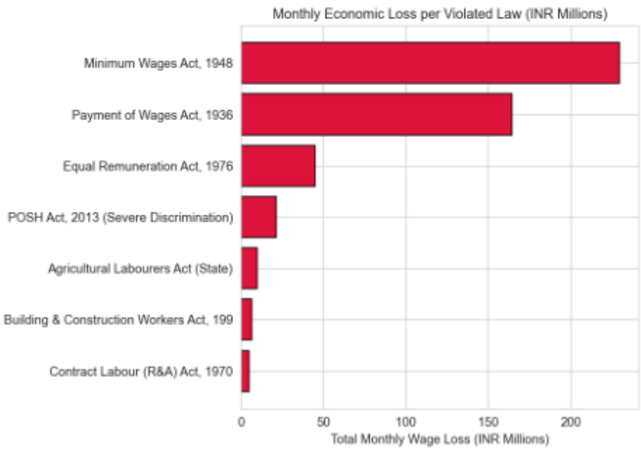

Fig. 11. Monthly Economic Loss due to Law Violations

VI. Conclusion

The study demonstrates that AI-driven models can effectively detect wage inefficiencies across Indian industries, with ensemble methods outperforming baseline approaches in capturing complex wage patterns. The results reveal significant underpayment, affecting over half the workforce, with agriculture and states like Uttar Pradesh showing the highest vulnerability, particularly among low-educated workers. Feature importance analysis highlights that wage disparities are primarily driven by structural factors such as actual wages, education and regional direct roles. The exploratory Quantum Machine Learning model shows potential in modelling complex interactions, reinforcing the value of advanced AI techniques for scalable, transparent and data-driven wage analysis, with scope for further refinement and real-world deployment.

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